		(1)/(2): $tan\ tan\ \theta = \frac{4\pi^2 r}{gT^2} \Rightarrow T = \sqrt{\frac{4\pi^2 r}{gtantan\ \theta}}$
10	**C**	Note that the satellites are *R* and 3*R* from the surface of Earth, not the centre of the Earth. Gravitational force provides for the centripetal force. $\frac{mv^2}{r} = \frac{GMm}{r^2} \Rightarrow v = \sqrt{\frac{GM}{r}} \Rightarrow v \propto \frac{1}{\sqrt{r}}$ $\frac{v_P}{v_Q} = \sqrt{\frac{r_Q}{r_P}} = \sqrt{\frac{4R}{2R}} = \sqrt{2}$ $\therefore v_P = \sqrt{2}v_Q = \sqrt{2}v$
11	**B**	$\phi = -\frac{GM}{r} \Rightarrow \phi \propto \frac{M}{r}$